

MAD & PWA Lab

Experiment No.	02
Experiment Title.	To design Flutter UI by including common widgets.
Roll No.	01
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Class	D15A
Subject	MAD & PWA Lab
Lab Outcome	LO2: Design and Develop interactive Flutter App by using widgets, layouts, gestures and animation
Grade:	

Experiment No. 2

Aim:-

To design Flutter UI by including common widgets.

Input Widgets:-

1. TextField: Allows text input.

```
TextField(  
  decoration: InputDecoration(labelText: 'Enter Name', border:  
    OutlineInputBorder()),  
)
```

2. Button Widgets:

a. ElevatedButton: Raised button with elevation.

```
ElevatedButton(  
  onPressed: () {}, child: Text('Click Me'),  
)
```

b. TextButton: Flat button with no elevation.

```
TextButton( onPressed: () {}, child: Text('Click Me'),  
)
```

c. IconButton: A button with an icon.

```
IconButton(  
  icon: Icon(Icons.add), onPressed: () {},  
)
```

D. Navigation & Routing Widgets

1. Navigator: Handles screen transitions.

```
Navigator.push(context, MaterialPageRoute(builder: (context) =>
```

```
  NewScreen())); 2. BottomNavigationBar: Used for switching between different  
screens. BottomNavigationBar(  
  items: [ BottomNavigationBarItem(icon: Icon(Icons.home), label:
```

```
    'Home'),  
    BottomNavigationBarItem(icon: Icon(Icons.settings), label: 'Settings'),  
  ],  
)
```

E. Interactive & Gestures GestureDetector: Detects taps, swipes, and other gestures. GestureDetector(
 onTap: () => print('Tapped!'),
 child: Container(color: Colors.blue, height: 100, width: 100),
)

)

Code:-*Profile*

```
import 'package:flutter/material.dart';
import 'package:flutter_svg/flutter_svg.dart';
import 'package:foodie/theme/text_styles.dart';

class ProfileTitleRow extends StatelessWidget {
  const ProfileTitleRow({
    super.key,
    required this.icon,
    required this.label,
  });

  final String icon;
  final String label;

  @override
  Widget build(BuildContext context) {
    return Padding(
      padding: const EdgeInsets.symmetric(vertical: 15),
      child: Row(
        mainAxisAlignment: MainAxisAlignment.spaceBetween,
        children: [
          SvgPicture.asset(icon),
          const SizedBox(
            width: 15,
          ),
          Expanded(
            child: Text(
```

```
        label,  
        style: TextStyles.titleProfile,  
      ),  
    ),  
  ],  
),  
);  
}  
}
```

Homepage

```
import 'package:flutter/material.dart';  
import 'package:flutter_riverpod/flutter_riverpod.dart';  
import 'package:foodie/router/router_context_extension.dart';  
import 'package:foodie/services/points/points.dart';  
import '../common/is_saved_material_button.dart';  
import '../domain/recipe.dart';  
import 'package:cached_network_image/cached_network_image.dart';  
import 'package:shimmer/shimmer.dart';  
import '../theme/theme.dart';
```

```
class RecipeWidget extends ConsumerWidget {  
  const RecipeWidget({  
    super.key,  
    required this.recipe,  
  });
```

```
  final Recipe recipe;
```

```
  @override
```

```
Widget build(BuildContext context, WidgetRef ref) {  
  return GestureDetector(  
    onTap: () => context.pushDetailPage(recipe: recipe),  
    child: Stack(  
      alignment: Alignment.bottomCenter,  
      children: [  
        Container(  
          height: 120,  
          alignment: Alignment.bottomCenter,  
          decoration: BoxDecoration(  
            borderRadius: BorderRadius.circular(  
              16,  
            ),  
            color: ThemeColors.main),  
        ),  
        Padding(  
          padding: const EdgeInsets.symmetric(horizontal: 15),  
          child: Column(  
            crossAxisAlignment: CrossAxisAlignment.start,  
            mainAxisAlignment: MainAxisAlignment.spaceAround,  
            children: [  
              Align(  
                alignment: Alignment.topCenter,  
                child: Stack(  
                  children: [  
                    ClipRRect(  
                      borderRadius: BorderRadius.circular(16),  
                      child: CachedNetworkImage(  
                        width: 150,  
                        height: 150,
```

```
        fit: BoxFit.cover,
        imageUrl: recipe.imageUrl,
        placeholder: (context, url) => Shimmer.fromColors(
          baseColor: Colors.black26,
          highlightColor: Colors.black12,
          child: Container(
            color: ThemeColors.white,
          ),
        ),
      ),
    ),
  ),
  Padding(
    padding: const EdgeInsets.all(10),
    child: IsSavedMaterialButton(
      recipe: recipe,
    ),
  ),
],
),
),
Text(
  recipe.name,
  maxLines: 2,
  style: TextStyles.recipeName,
),
Padding(
  padding: const EdgeInsets.only(bottom: 5),
  child: Row(
    mainAxisAlignment: MainAxisAlignment.spaceBetween,
    children: [
```

```
        Text(
          '${Points.recipePoints} points',
          style: TextStyles.points,
        ),
        Text(
          '${recipe.time != null ? (recipe.time != 0 ? recipe.time : "") : ""}
          ${recipe.time != null ? (recipe.time != 0 ? 'mins' : "") : ""}',
          style: TextStyles.time,
        ),
      ],
    ),
  ),
],
),
)
],
),
)
),
);
}
}
```

Saved

```
import 'package:cached_network_image/cached_network_image.dart';
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';
import 'package:foodie/constants/storage_box_constants.dart';
import 'package:foodie/providers/providers.dart';
import 'package:foodie/router/router_context_extension.dart';
import 'package:shimmer/shimmer.dart';
import '../constants/string_constants.dart';
```

```
import '../services/points/points.dart';
import '../theme/theme.dart';
import '../domain/recipe.dart';

class SavedRecipesPage extends ConsumerWidget {
  const SavedRecipesPage({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context, WidgetRef ref) {
    final storageService = ref.watch(storageServiceProvider);
    final imageSize = MediaQuery.of(context).size.width * 0.3;
    final favoriteBoxSize =
      storageService.getLength<Recipe>(StorageBox.favoritesBox);
    return WillPopScope(
      onWillPop: () async => false,
      child: Scaffold(
        appBar: AppBar(
          automaticallyImplyLeading: false,
          title: Text(
            StringConstants.savedRecipes,
            style: TextStyles.title,
          ),
        ),
        body: SafeArea(
          child: favoriteBoxSize == 0
            ? Center(
                heightFactor: MediaQuery.sizeOf(context).height * 0.5,
                child: Text(
                  StringConstants.noSavedRecipes,
                  style: TextStyles.subtitle,
```



```
    ),  
  )  
: Column(  
  children: [  
    Expanded(  
      child: ListView.builder(  
        shrinkWrap: true,  
        itemCount: favoriteBoxSize,  
        itemBuilder: (context, index) {  
          final recipe = storageService  
            .getAll<Recipe>(StorageBox.favoritesBox)[index];  
          return Padding(  
            padding: const EdgeInsets.fromLTRB(20, 20, 20, 0),  
            child: GestureDetector(  
              onTap: () =>  
                context.pushDetailPage(recipe: recipe),  
              child: Row(  
                children: [  
                  ClipRRect(  
                    borderRadius: BorderRadius.circular(16),  
                    child: CachedNetworkImage(  
                      width: imageSize,  
                      height: imageSize,  
                      fit: BoxFit.cover,  
                      imageUrl: recipe.imageUrl,  
                      placeholder: (context, url) =>  
                        Shimmer.fromColors(  
                          baseColor: Colors.black26,  
                          highlightColor: Colors.black12,  
                          child: Container(  
                            width: 100%,  
                            height: 100%,  
                            decoration: BoxDecoration(  
                              color: Colors.white,  
                              borderRadius: BorderRadius.circular(16),  
                            ),  
                          ),  
                        ),  
                    ),  
                  ),  
                ],  
              ),  
            ),  
          ),  
        ],  
      ),  
    ),  
  ],  
)
```

```
        color: ThemeColors.white,
      ),
    ),
  ),
),
const SizedBox(
  width: 20,
),
Expanded(
  child: Column(
    crossAxisAlignment:
      CrossAxisAlignment.start,
    children: [
      Text(
        recipe.name,
        style: TextStyles.title,
      ),
      const SizedBox(
        height: 20,
      ),
      Text('${Points.recipePoints} points',
        style: TextStyles.points)
    ],
  ),
),
],
),
),
);
}},
```

```
        )  
    ],  
    ),  
    ),  
    ),  
    );  
}  
}
```

Output:-





